

Week 1 Assignment – E-commerce

EDA Scenario Challenge

Phase 1 – Dataset Examination

The dataset contains approximately 500,000 observations and 30 attributes comprising numerical, categorical, binary, and date-related data. Several columns, including `return_reason`, `shipping_cost`, `days_to_deliver`, `customer_age`, `customer_lifetime_value`, `rating`, `pages_viewed`, and `session_duration_mins`, contain missing values that must be addressed during preprocessing. Duplicate records should also be identified and removed. Furthermore, the `order_date` field should be converted into a datetime data type for effective analysis.

Phase 2 – Analysis of Pricing Variables

The pricing features `product_base_price`, `final_price`, and `total_amount` exhibit a right-skewed distribution. Visualizations such as histograms reveal that most transactions fall within lower and moderate price ranges, while a relatively small number of costly purchases contribute to the long tail on the higher end of the distribution.

Phase 3 – Evaluation of Categorical Features

Several categorical attributes are present in the dataset, including `category`, `sub_category`, `shipping_method`, `payment_method`, `customer_gender`, `country`, `city`, and `loyalty_tier`. Features with numerous unique values, particularly `city`, should be managed carefully during encoding. Since `loyalty_tier` follows a natural ranking, it should be encoded in the order:

Bronze < Silver < Gold < Platinum

Phase 4 – Return Behavior Investigation

The variable `is_returned` serves as the target for analyzing product return patterns. Factors such as customer ratings, delivery performance, discount levels, loyalty status, and product category may significantly influence return decisions. The `return_reason` attribute is meaningful only for returned items and therefore is expected to have null values for orders that were not returned.

Phase 5 – Data Preparation and Feature Engineering

A preprocessing framework can be implemented using **ColumnTransformer**. Missing numerical values should be replaced using median imputation and standardized using **StandardScaler**. For highly

skewed numerical variables such as **total_amount**, **RobustScaler** can provide better normalization. Nominal categorical attributes should be transformed using **One-Hot Encoding**, while the ordinal **loyalty_tier** feature should be encoded according to its hierarchy.

Phase 6 – Relationship and Correlation Analysis

Correlation analysis should be conducted on numerical variables to uncover relationships among features and identify potential multicollinearity issues. Variables including **days_to_deliver**, **rating**, **discount_percentage**, **customer_lifetime_value**, and customer purchasing metrics may have a strong impact on predicting whether an order will be returned, making them valuable inputs for predictive modeling.

Phase 6 – Correlation Analysis

Correlation analysis should be performed on numerical variables to identify multicollinearity and relationships with the target variable `is_returned`. Features such as `days_to_deliver`, `rating`, `discount_percentage`, `customer_lifetime_value`, and purchase behavior metrics may provide useful predictive information for future return prediction models.